John Ian R. So

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Education

University of California, Berkeley **GPA: 4.00 / 4.00**

B.S. Electrical Engineering and Computer Science, May 2023

Courses Completed: Structure & Interpretation of Computer Programs, Data Structures, Machine Structures, Discrete Mathematics & Probability Theory, Designing Information Devices & Systems I, Self-Driving Cars, Multivariable Calculus, Physics for Scientists & Engineers.

Courses in Progress: Efficient Algorithms & Intractable Problems, Designing Information Devices & Systems II.

Honors: Eta Kappa Nu (Mu Chapter, inducted April 2020).

Work Experience

Robotics Intern, PhaseSpace Motion Capture *May-August 2020*

- Implemented robot system architecture in ROS Melodic for a SLAM navigation stack using depth camera and IMU odometry. Modelled a robot frame using URDF, and simulated sensors and actuators in Gazebo.
- Researched high contact surface sanitation as a response to COVID-19.
- Prototyped software, firmware, and hardware using an ESP32 controller and Firebase for the web interfaced, closed loop control system of a hypochlorous acid sanitizer generator and dispenser system.

Lead Intern, PhaseSpace Motion Capture *June-August 2018*

- Led team of interns in building a robot arm driven by an ESP32 controller and PhaseSpace's motion capture system.
- Created the arm's control system in Arduino to translate 3D vectors to arm joint angles.
- Developed software in Java to translate motion capture data to scaled vectors and send representations to the arm control system.

Activities

Member, Machine Learning at Berkeley *September 2020-Present*

Research Assistant, Biomimetic Millisystems Lab *September 2020-Present*

- Beginning research under Professor Ronald Fearing on cooperative multi-robot interactions and control.
- Simulating algorithms using CoppeliaSim, Kamigami robots, and ROS.

Course Staff, CS 61A *August 2020-Present*

- Leading discussion sections four times per week and holding office hours twice per week covering functional, object-oriented, and declarative programming in Python, Scheme, and SQL for CS 61A: *Structure and Interpretation of Computer Programs*, UC Berkeley's flagship CS course of over 1500 students.
- Developing software tools for staff use, including Slack bot integration.

CS 61A Mentor, Computer Science Mentors *January 2020-Present*

EECS Engineer, Formula Electric at Berkeley *January 2020-Present*

- Engineered PCB designs in Altium Designer for the tractive system indicator and driver interface subsystems of an electric vehicle for the Formula SAE Competition EV division.
- Creating firmware for STM controllers using Mbed OS.

Skills

Languages: Java, C, C++, JavaScript, Python, RISC-V, SQL, HTML/CSS, Ruby.
Packages/Frameworks: Flask, React, Android Studio, OpenMP, Node.js.
Robotics: ROS, OpenCV, Keras, TensorFlow, Arduino, ESP.
Tools: Git, Unix Shell, IntelliJ.
Circuit/PCB Design: Altium Designer, Multisim, Autodesk Eagle.

Projects

Mask Vision *August 2020-Present*

- Trained a CNN and automated data augmentation using OpenCV and Keras to classify masks.
- Currently looking into transfer learning using YOLO for live recognition of masks in video.

NumC *July-August 2020*

- Implemented a matrix library in C and as a Python module using a Python-C interface.
- Optimized operations using Intel SSE Intrinsics, OpenMP, loop unrolling, and cache blocking. Achieved ~2400x speedup against unoptimized benchmarks.

Word Hunt Solver *May-August 2020*

- Implemented a model-view-controller program for Word Hunt, a game where players swipe words on a 4x4 board, and a solver script to find all possible words in one second using DFS graph traversal.
- Integrated the solver and model into a full-stack web application using Flask and ReactJS.

Self-Driving Car Simulation *2019*

- Created models, including PID, regression and neural networks, for self-driving car subsystems including acceleration control, route planning, and steering control using TensorFlow, OpenCV, and Duckietown.
- Used PID, A*, and cost maps to plan and run routes.